

WEEK 8 ASSIGNMENT
H524 Introduction to Biostatistics - Fall 2025

Total Points: 10 — **Due:** Sunday 11:59 PM

Part A: Multiple Choice Questions (9 points)

Answer the following 9 questions on contingency tables and chi-square tests. Each question is worth 1 point.

Question 1: Contingency Table Basics (1 point)

A contingency table displays the relationship between:

- A. One categorical variable
- B. Two continuous variables
- C. Two categorical variables
- D. One categorical and one continuous variable

Question 2: Expected Frequencies (1 point)

In a chi-square test, the expected frequency for a cell is calculated as:

- A. $(\text{Row total} \times \text{Column total}) / \text{Grand total}$
- B. $(\text{Row total} + \text{Column total}) / 2$
- C. $\text{Grand total} / \text{Number of cells}$
- D. $\text{Observed frequency} / \text{Total sample size}$

Question 3: Chi-Square Test Assumptions (1 point)

Which of the following is a requirement for conducting a chi-square test?

- A. All expected cell frequencies must be at least 10
- B. All expected cell frequencies should be at least 5
- C. The sample must be normally distributed
- D. Variables must be continuous

Question 4: Degrees of Freedom (1 point)

For a 3×4 contingency table (3 rows, 4 columns), what are the degrees of freedom for the chi-square test?

- A. 7
- B. 6
- C. 12
- D. 11

Question 5: Interpreting Chi-Square Results (1 point)

A chi-square test yields $p = 0.032$. At $\alpha = 0.05$, what can we conclude?

- A. There is no association between the variables
- B. There is a significant association between the variables
- C. The variables are normally distributed
- D. The test assumptions are violated

Question 6: Relative Risk (1 point)

In a study of disease and exposure, the relative risk (RR) is:

- A. The ratio of the odds of disease in exposed vs. unexposed groups
- B. The ratio of the risk of disease in exposed vs. unexposed groups
- C. The difference in risk between exposed and unexposed groups
- D. Always equal to 1 if there is no association

Question 7: Odds Ratio (1 point)

An odds ratio of 2.5 means:

- A. The exposed group has 2.5 times the risk of disease compared to unexposed
- B. The odds of disease in the exposed group are 2.5 times the odds in the unexposed group
- C. There is a 2.5% difference in disease rates between groups
- D. The chi-square statistic is 2.5

Question 8: 2×2 Table Setup (1 point)

In a case-control study examining smoking and lung cancer, which variable should form the rows of a 2×2 table?

- A. Smoking status (exposed/unexposed)
- B. Lung cancer status (case/control)
- C. Either variable can form rows
- D. Neither - case-control studies cannot use 2×2 tables

Question 9: Fisher's Exact Test (1 point)

When should Fisher's exact test be used instead of the chi-square test?

- A. When sample size is very large ($n > 1000$)
- B. When expected cell frequencies are small (less than 5 in one or more cells)
- C. When variables are continuous
- D. When there are more than two categories

Part B: Group Project Check-In - Progress Update (1 point)

GROUP PROJECT PROGRESS CHECK

Reminder: This Wednesday (12:00-1:20pm) is **Office Hours Day** - drop by classroom/office if your group needs help!

Your Assignment

Group Leader submits a brief progress update:

Answer these questions:

1. **Has your group met at least once since Week 7?**
☐ Yes
☐ No
2. **Have you loaded and explored your dataset in R?**
☐ Yes
☐ No
3. **Have you consulted at least one AI tool about your data?**
☐ Yes (which one(s)? _____)
☐ No
4. **Any blockers or questions?** (optional)

Note: This week is Office Hours Day! If you answered "No" to any question above or have blockers, come to Wednesday's class time (12-1:20pm) for help.

Submission Format

Submit as: Text entry (Canvas quiz-style) OR brief paragraph

Length: 3-5 sentences

Grading: 1 point for completion (just answer honestly!)

Notes

- **Only the group leader submits**
- This is a quick check-in (1 point) - not a full report
- If you're stuck, **come to Office Hours Day Wednesday 12-1:20pm**
 - Location: CORD 1424 (classroom) 12:00-12:30, then 153 Milam (office) 12:30-1:20
 - No appointment needed, just drop by
 - Bring your laptop and dataset
- Groups are expected to be working on analysis this week